

131. $a + b = 2c \Rightarrow a = 2c - b$.

$$\begin{aligned}\therefore \frac{a}{a-c} + \frac{b}{b-c} &= \frac{2c-b}{(2c-b)-c} + \frac{b}{(b-c)} = \frac{2c-b}{c-b} + \frac{b}{b-c} \\ &= \frac{b-2c}{b-c} + \frac{b}{b-c} = \frac{b-2c+b}{b-c} = \frac{2(b-c)}{b-c} = 2.\end{aligned}$$

132. $\frac{x}{x-1} + \frac{1}{x+1} + \frac{2x}{1-x^2} = \frac{x}{x-1} + \frac{1}{x+1} - \frac{2x}{x^2-1}$

$$\begin{aligned}&= \frac{x(x+1) + (x-1) - 2x}{(x^2-1)} \\ &= \frac{x^2 + x + x - 1 - 2x}{x^2-1} = \frac{x^2-1}{x^2-1} = 1.\end{aligned}$$

133. $\frac{(a+b)^2}{(a^2-b^2)} = \frac{(a+b)(a+b)}{(a+b)(a-b)} = \frac{a+b}{a-b}$.

134. $ab + bc + ca = 0 \Rightarrow ab = -bc - ca, bc = -ab - ca, ca = -ab - bc$.

$$\begin{aligned}\therefore \frac{1}{a^2-bc} + \frac{1}{b^2-ca} + \frac{1}{c^2-ab} \\ &= \frac{1}{a^2+ab+ac} + \frac{1}{b^2+ab+bc} + \frac{1}{c^2+bc+ca} \\ &= \frac{1}{a(a+b+c)} + \frac{1}{b(a+b+c)} + \frac{1}{c(a+b+c)} \\ &= \frac{bc+ca+ab}{abc(a+b+c)} = 0. [\because ab+bc+ca=0]\end{aligned}$$

135. Number of boxes required = $\frac{15 \times 1000}{250} = 60$.

136. Bus fare for 1 person = ₹ 420.
Train fare for 1 person = ₹ $\left(\frac{3}{4} \times 840\right)$ = ₹ 630.

$$\therefore \text{Total fare} = ₹ (3 \times 420 + 4 \times 630) = ₹ (1260 + 2520) = ₹ 3780.$$

137. Let the cost of one pen be ₹ x . Then, cost of 1 pencil = ₹ $\left(\frac{x}{3}\right)$.

$$\therefore 6x + 3 \times \frac{x}{3} = 84 \Rightarrow 7x = 84 \Rightarrow x = 12.$$

So, total cost of 4 pens and 5 pencils

$$= ₹ \left(4x + \frac{5x}{3}\right) = ₹ \left(\frac{17x}{3}\right) = ₹ \left(\frac{17 \times 12}{3}\right) = ₹ 68.$$

138. Amount received by each person = ₹ $\left(\frac{436563}{69}\right)$ = ₹ 6327.

$$\begin{array}{r} 69 \overline{)436563} \quad 6327 \\ \underline{414} \\ 225 \\ \underline{207} \\ 186 \\ \underline{138} \\ 483 \\ \underline{483} \\ \times \end{array}$$

139. Length of reel = $(32 \times 120 + 80)$ cm = 3920 cm = 39.20 m.

140. Number of bananas required daily = $\frac{798}{7} = 114$.

$$\begin{aligned}\therefore \text{Required number} &= [114 \times (\overset{\text{Jan}}{31} + \overset{\text{Feb}}{29} + \overset{\text{Mar}}{31})] \\ &= (114 \times 91) = 10374.\end{aligned}$$

141. Suppose Mohan has ₹ x . Then, Ram has ₹ $(x + 6)$ and Sohan has ₹ $(x - 3)$.

$$\begin{aligned}\therefore x + (x + 6) + (x - 3) &= 33 \Leftrightarrow 3x = 30 \Leftrightarrow x = 10. \\ \text{So, Ram's share} &= ₹ (10 + 6) = ₹ 16.\end{aligned}$$

142. 4 gal = (4×4) qt = 16 qt = (16×2) pt = 32 pt.

$$\therefore \text{Number of bottles} = \frac{32}{\left(\frac{1}{2}\right)} = 64.$$

143. $\frac{1}{2}A = \frac{5}{6}B \Rightarrow A = \left(2 \times \frac{5}{6}\right)B = \frac{5}{3}B$.

$$\begin{aligned}\therefore A + B = 80 &\Rightarrow \frac{5}{3}B + B = 80 \Rightarrow \frac{8}{3}B = 80 \\ &\Rightarrow B = \left(\frac{80 \times 3}{8}\right) = 30 \text{ kg}.\end{aligned}$$

144. $a = b + 2 \Rightarrow b = a - 2$

$$\begin{aligned}\text{And, } b = c + 10 &\Rightarrow c = b - 10 = (a - 2) - 10 = a - 12. \\ a + b + c = 130 &\Rightarrow a + (a - 2) + (a - 12) = 130 \\ &\Rightarrow 3a = 144 \Rightarrow a = 48. \\ \therefore (b + c) - a &= [(a - 2) + (a - 12)] - a \\ &= (2a - 14) - a = (a - 14) = 48 - 14 = 34.\end{aligned}$$

145. Let the given condition be fulfilled n years after 2002.

$$\begin{aligned}\text{Then, price of item X after } n \text{ years} &= ₹ (420 + 40n). \\ \text{price of item Y after } n \text{ years} &= ₹ (630 + 15n). \\ \therefore 420 + 40n &= (630 + 15n) + 40 \\ \Rightarrow 420 + 40n &= 670 + 15n \Rightarrow 25n = 250 \Rightarrow n = 10. \\ \text{So, the required year is } &10 \text{ years after 2002 i.e., 2012.}\end{aligned}$$

146. Number of pieces = $\left(\frac{42.5 \times 100}{85}\right) = \frac{4250}{85} = 50$.

147. Total number of children = $(30 \times 16) = 480$.

$$\therefore \text{Number of columns of 24 children each} = \left(\frac{480}{24}\right) = 20.$$

148. Original number of sections = $(16 - 3) = 13$.

Original number of students = $(24 \times 13) = 312$.

Present number of students = $(21 \times 16) = 336$.

Number of new students admitted = $(336 - 312) = 24$.

149. Money earned in 2 days = ₹ $(20 - 15) = ₹ 5$.

$$\text{Money earned in 16 days} = ₹ \left(\frac{5}{2} \times 16\right) = ₹ 40.$$

On 17th day, money in hand = ₹ $(40 + 20) = ₹ 60$.

150. Let the number of cars sold on the first day be x .

$$\begin{aligned}\text{Then, } x + (x + 6) + (x + 12) + (x + 18) + (x + 24) \\ + (x + 30) &= 150 \\ \Leftrightarrow 6x + 90 &= 150 \Leftrightarrow 6x = 60 \Leftrightarrow x = 10.\end{aligned}$$

$$\therefore \text{Number of cars sold on 6th day} = (10 + 30) = 40.$$

151. Let the number of captains in the group be x .

Then, number of soldiers = $15x$.

$$\therefore x + 15x = 1200 \Leftrightarrow 16x = 1200 \Leftrightarrow x = 75.$$

152. Required cost = ₹ $[1000 \times x + (z - 1000) \times y]$

$$\begin{aligned}&= ₹ (1000x + zy - 1000y) \\ &= ₹ [1000(x - y) + yz].\end{aligned}$$